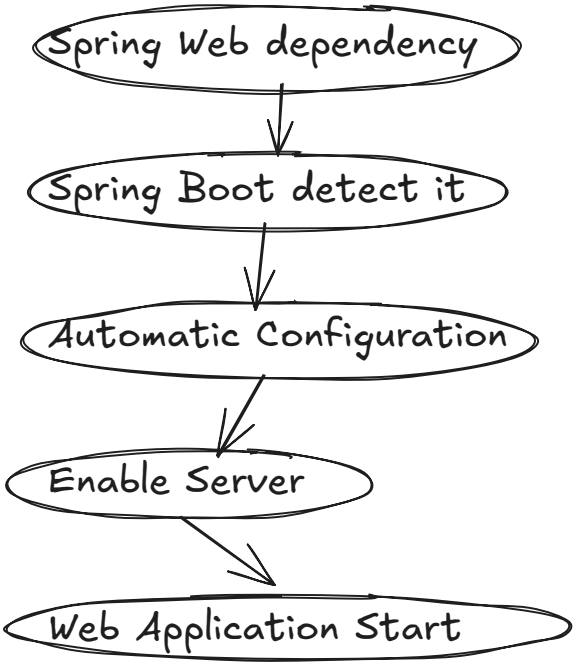
@SpringBootApplication
-> @SpringBootConfiguration
-> @EnableAutoConfiguration
-> @ComponentScan

- @SpringBootApplication
This is the main class of Spring Boot Application
Start the application from here and automatically configure the required components.

- @SpringBootConfiguration
It tells spring that this class contains the configuration for the application.

- @EnableAutoConfiguration
Automatically configure the application based on the dependency available in the project.

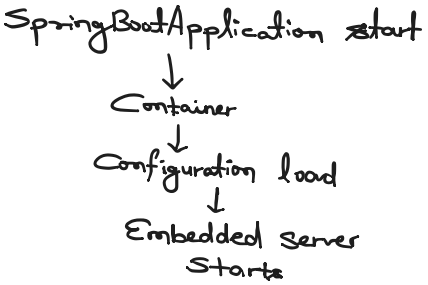
```
<dependency>
<group-id> org.springframework.boot </group-d>
<artifactId> spring-boot-starter-web </artifactId>
</dependency>
```



-> @ComponentScan

It tells the spring to find spring components in the application package.





--> @Configuration

@Configuration is used on a Java class that contains bean definition.

```
@Configuration
 public class MyConfig{
          }
```

--> @Bean

@Bean is used on a method.

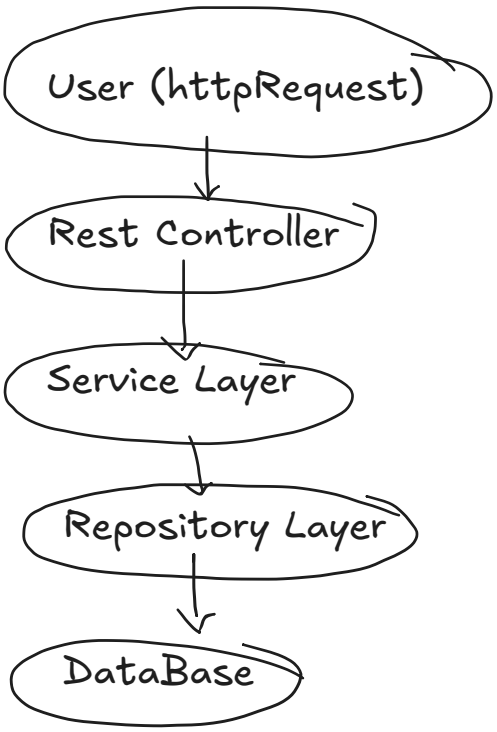
Note: Bean internal call the method, take the return object and register it in the Spring IOC container.

REST API

- It stands for Representational state transfer application programming interface.

- In a Spring Boot, Rest API are commonly used to allow a frontend, mobile app, or another application to communicate with the backend using HTTP.

ARCHITECTURE OF SPRING BOOT REST API



Rest API Http Method

1. Get - Read/fetch Data
   used to retrieve data from the server

   Request : Get /api/accounts/101

   Response : { "account_id" : 101,
   "name" : "Raunak",
   "balance": 21273
   }

2. Post (Create New Data)
   used to create a new resource

   Request : { "name" : "Raunak",
   "balance" : 151
   }
   Response : { "account_id" : 102,
   "name" : "Raunak",
   "balance" : 151
   }

3. PUT (Update Complete Data)
used to completely update an existing resource.

Request : PUT /api/accounts/102

```
{"name":"Prachi",
   "balance":101
    }
```

4. PATCH (Update Partial Data)
used when you want to update only some fields

Request : PATCH /api/accounts/102

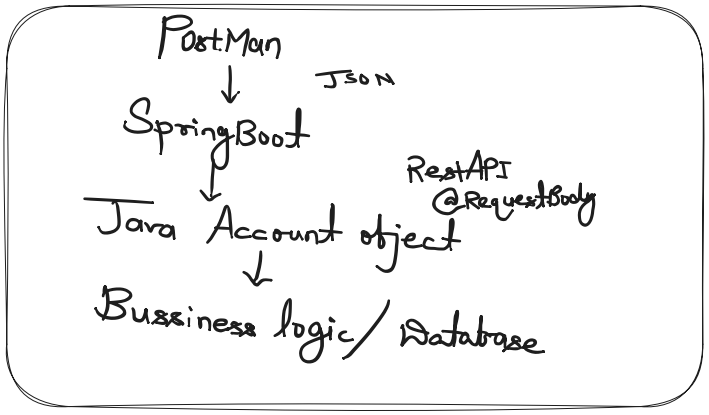
```
{"name":"Ram"}
```

5. DELETE (Delete Data)
used to remove a resource

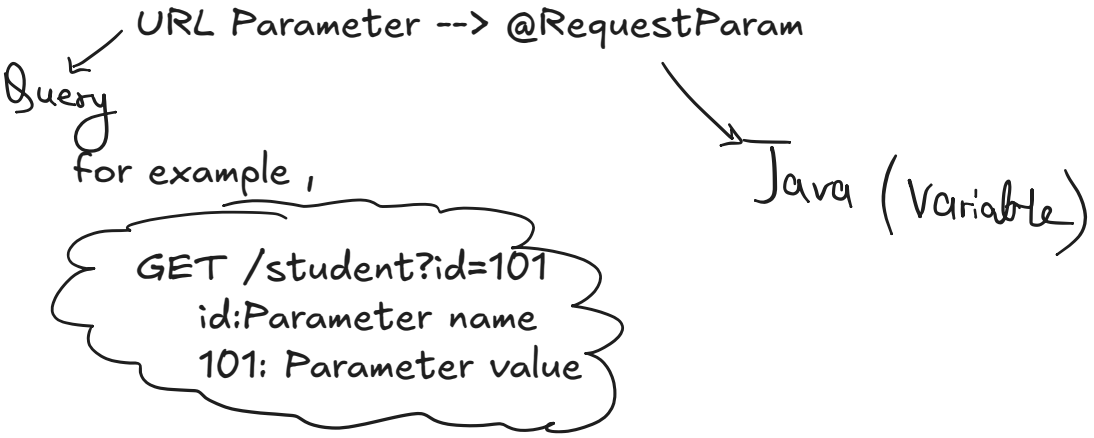
Request : DELETE /api/account/102

@Autowired

@Autowired is used in SpringBoot for dependency injection.
It tells spring -
" Find the required object (Bean) & automatically provide it here. "



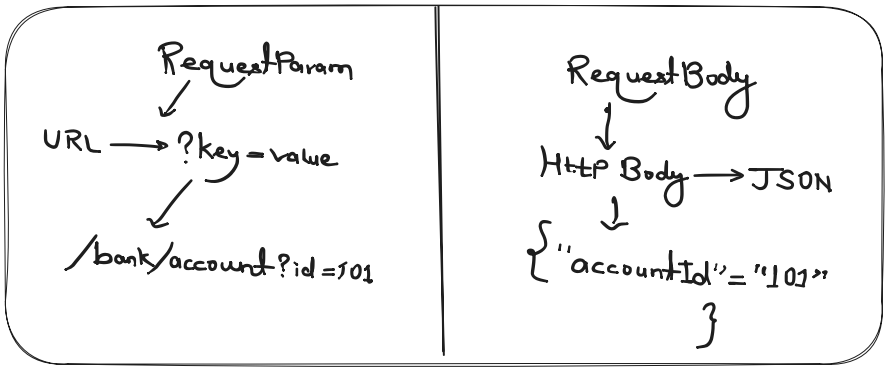
@RequestParam is used to read value from the URL, Query Parameter in springboot Rest API.



What is the difference between RequestBody and RequestParam

RequestBody	RequestParam
--> Reads data from JSON Body. --> Commonly used. --> Used with POST, PUT, PATCH -->Convert JSON into Java Object	--> Reads data from query parameter. --> Commonly used for small parameters. --> Often used with GET --> Convert Param into Java Value

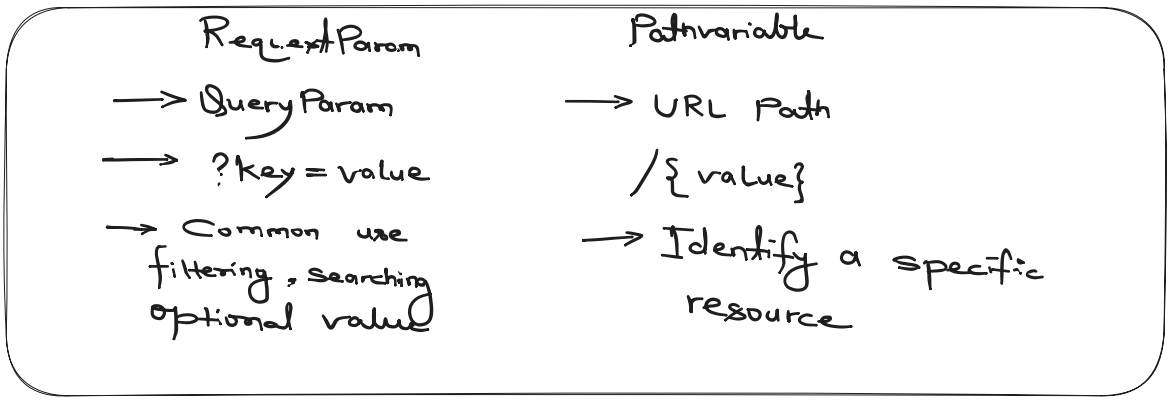
Easy way understanding



@PathVariable

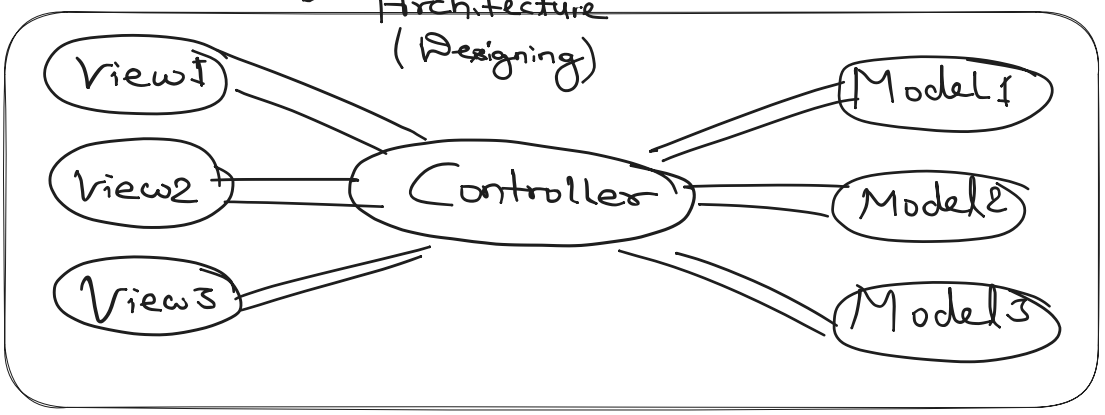
It is use to extract a value from the URL path and use it inside a controller method.

--> Difference between RequestParam and PathVariable



MVC (Model-View-Controller)

→ Architecture (Designing)



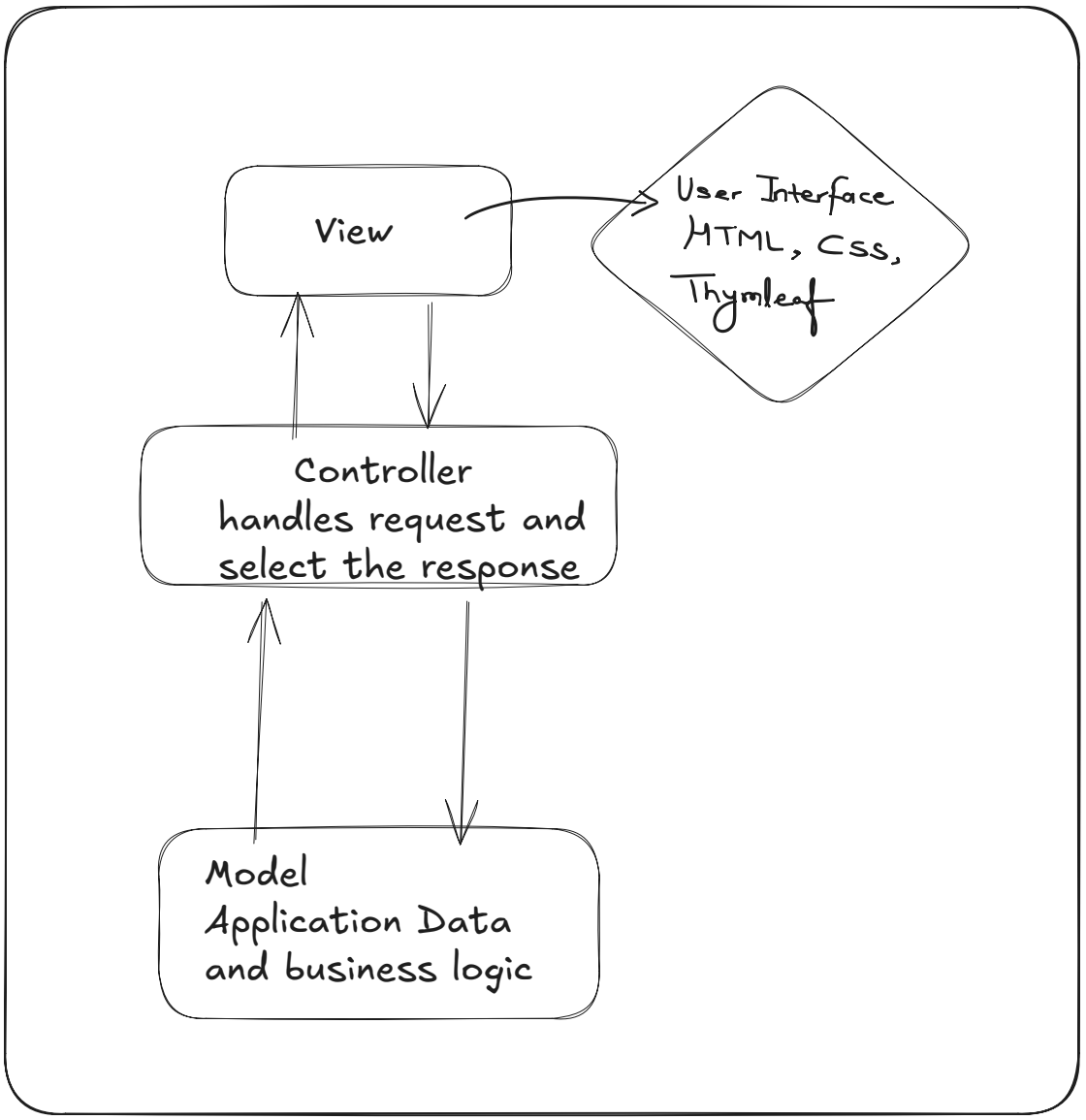
#loosely Coupling Application
#Maintenance easier

MVC

It is a software design pattern use to separate on application data, user interface and business request handling logic.

- In Spring Boot MVC, It is commonly used to develop web application such as enterprise application.

--> Architecture of MVC
MVC has three main component



1. Model :
   -> Represents application data
   -> Contains data objects and may work with business or service-layer logic.
   -> Often uses entities, DTO (Data transfer object) and repositories in Spring Boot Application.

2. View :
   -> Display information to the user.
   -> Common technologies include HTML, CSS, Javascript and Thymleaf.

3. Controller :
   -> Receive HTTP request from the user.
   -> Calls the service layer when business logic is needed.
   -> Return a view name or an HTTP response.

How MVC work in SpringBoot

Consider a E-Commerce System

Step 1 : User
Step 2 : Dispatcher Servlet
Step 3 : Controller
Step 4 : Service/Repository
Step 5 : View

Task : Create a spring boot web application
with frontend using thymleaf in which a page accepts
Name of student
course of student
and marks
and shows the input with conditions in another page or same
marks > 90 --> A grade
marks < 90 --> B grade
marks <50 --> Fail